Jay Raval

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Education

Ahmedabad University **Aug 2022 – May 2026**
Bachelor of Technology in Computer Science and Engineering *CGPA: 3.5/4.00*

Experience

Sourceasy.ai **May 2025 – Jul 2025**
AI Intern *Ahmedabad, Onsite*

- Developed India's first AI-powered B2B chemical sourcing platform, streamlining procurement with intelligent supplier-buyer matching and real-time analytics.
- Built a WhatsApp-based AI agent for multilingual request parsing, automating quote requests to suppliers.
- Integrated Pinecone(VectorDB) for data storage and similarity search, and supplier matching.
- Designed a responsive React-based UI with Framer Motion and GSAP animations, supported by a modular microservices architecture.
- **Technologies Used:** React, TypeScript, Node.js, Express, Python, Pinecone, Firebase, Chart.js, Framer Motion, GSAP, Tailwind CSS, Render, Redis, Twilio

Papers Under Publication

Testing the Limits of Large Language Models on Regular Languages
Jay Raval, Shlok Shelat, Souvik Roy, Manas Gaur *Under Review at EMNLP 2026*, May 2026

Beyond Memorization: Testing LLM Reasoning on Unseen Theory of Computation Tasks
Shlok Shelat, Jay Raval, Souvik Roy, Manas Gaur *Accepted ACL Findngs 2026*, January 2026

Is it possible to predict phase transition in cellular automata?
Jay Raval, Aarya Parekh, Sumit Adak, Souvik RoyUnder Review at the *International Journal of Bifurcation and Chaos*, April 2026

Understanding the Game of Collaboration through H-Index
Jinil Savaj, Jay Raval, Amit A. Nanavati, Souvik Roy Under Revision

Projects

Hybrid P2P - Content Delivery Network | *GO, Docker, Redis, PostgreSQL* **Sep 2025 – Nov 2025**

- Designed and implemented a hybrid Peer-to-Peer and CDN-based content delivery architecture to reduce server load and improve scalability by enabling clients to share content segments dynamically.
- Developed intelligent content distribution and peer selection mechanisms to optimize latency, bandwidth utilization, and fault tolerance under varying network conditions.
- **Technologies Used:** Go, Docker, Redis, PostgreSQL, Graph Algorithms

Identifying Hard Stop & Momentary Stop Using Vehicle Trajectory Data | *Python, scikit-learn, ML* **Jan 2025 – Apr 2025**

- Classified vehicle behavior at intersections into moving, momentary stop, and hard stop using aerial-view trajectory datasets for traffic analysis and road safety.
- Successfully tested the system on multiple collected datasets, demonstrating strong generalization across varied environmental and sampling conditions.
- **Technologies Used:** Python, scikit-learn, Random Forest, K-Means, HMM, pandas, NumPy, matplotlib

Digivote - Blockchain Voting System | *Go, Solidity, Tendermint, Ethereum* **Jan 2025 – Apr 2025**

- Developed a secure, decentralized voting platform using a hybrid blockchain with Tendermint ABCI and Ethereum smart contracts, implementing DPoS consensus.
- Built cross-chain synchronization, secure validator management, and a terminal-based dashboard with Bubble Tea for real-time proposal tracking.
- **Technologies Used:** Go, Solidity, Tendermint, Ethereum, Truffle, Ganache, Bubble Tea, GoLevelDB, SHA-256

Read.AI - Context-Aware Study Assistant | *React, Node.js, Express, LLAMA, Vector Databases* **Oct 2024 – Jan 2025**

- Built an AI-powered study tool parsing PDFs into vector embeddings for a vector database, enabling context-aware Q&A with LLAMA model and cosine similarity search.
- Implemented Study Mode for auto-generated flashcards and quizzes to enhance exam preparation.
- **Technologies Used:** React, Node.js, Express, LLAMA, Vector Databases, Cosine Similarity

Advanced Rijndael/AES Encryption System | *C++, STL, Threading* **Aug 2023 – Nov 2023**

- Developed a robust AES encryption system in C++ supporting 128/192/256-bit keys and ECB/CBC/CTR modes for secure data encryption.
- Implemented core AES transformations (SubBytes, ShiftRows, MixColumns, AddRoundKey) with secure key generation and thread-safe operations.
- **Technologies Used:** C++, STL, Linear Algebra

Facial Detection Using Image Segmentation | *MATLAB, Image Processing, Machine Learning* **Aug 2023 – Nov 2023**

- Designed a custom algorithm for detecting human faces and features in images using colour segmentation, isolating facial regions (eyes, nose, mouth) via skin-tone clustering.
- Achieved facial feature extraction using k-means clustering in LAB color space.
- **Technologies Used:** MATLAB, Image Segmentation, Image Processing, Machine Learning, Computer Vision

Certifications

IBM Data Science Orientation, Coursera Issued Jun 2024, *Skills:* Python, Data Analysis, Credential: PWID-B0267600

Red Hat Certified System Administrator (RHCSA) Issued Jan 2024, *Skills:* Red Hat Linux, System Administration, Credential ID: 240-001-560

Technical Skills

Languages: Python, C++, JavaScript, TypeScript, Go, MATLAB
Frameworks/Libraries/Tools: React, Node.js, Express, scikit-learn, Hugging Face Transformers, TensorFlow, Tailwind CSS, Chart.js, GSAP, Framer Motion, Git, Figma, Postman, Docker
Databases: MySQL, MongoDB, Pinecone, SQLite, Redis, PostgreSQL
DevOps & OS: RHEL 9, Linux Shell Scripting, Vercel, Render

Leadership / Extracurricular

CSE 408-Software Engineering, Ahmedabad University **Jan 2026 – May 2026**
Teaching Assistant *Ahmedabad*

- Mentoring student teams on designing, developing, testing, and deploying software applications.

ACM Chapter, Ahmedabad University **Oct 2024 – Nov 2025**
Vice Chair *Ahmedabad*

- Led organization of programming competitions, workshops to foster coding culture among students.